

DECATUR, TX, USA

WMO: 720305

Lat: 33.254N

Lon: 97.581W

Elev: 1047

StdP: 14.15

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53964

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.1	26.3	6.8	8.1	29.8	10.4	9.7	35.9	23.1	53.1	20.7	52.3	7.2	0	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.4	99.6	70.9	97.4	71.7	95.1	71.9	76.5	89.2	75.4	87.9	74.7	86.9	8.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	128.8	80.2	72.4	124.8	79.8	71.6	121.1	79.4	40.6	89.2	39.6	87.9	38.8	86.8	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	19.0	17.4	15.2	102.5	4.4	2.5	12.0	104.2	9.5	105.7	7.0	107.1	3.8	108.9
			13.6	77.7	3.8	1.8	10.8	79.0	8.6	80.0	6.5	81.0	3.7	82.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	45.8	49.1	57.3	64.2	71.7	80.4	83.7	83.9	77.1	66.1	55.9	46.4
	DBStd	15.93	10.23	11.10	10.02	7.99	6.84	4.61	4.62	5.23	6.02	8.71	9.36	9.56
	HDD50	644	205	142	49	7	0	0	0	0	0	6	49	186
	HDD65	2419	597	453	271	107	24	0	0	0	2	93	294	578
	CDD50	6202	75	118	276	433	674	912	1045	1050	814	506	225	74
	CDD65	2499	2	8	33	83	232	462	580	585	366	128	19	1
	CDH74	27413	21	64	231	615	1984	5127	7232	7317	3647	1039	123	13
	CDH80	13049	1	11	48	146	672	2402	3864	3984	1594	306	21	0
Wind	WSAvg	8.4	8.3	9.0	10.2	10.2	9.1	8.7	7.5	7.0	6.6	7.7	8.3	8.0
Precipitation	PrecAvg	37.2	1.7	2.5	3.4	3.5	4.5	4.3	2.5	2.1	3.6	4.0	2.5	2.6
	PrecMax	51.4	4.0	7.5	7.9	11.5	9.7	11.8	5.7	6.2	10.2	9.4	6.4	8.7
	PrecMin	24.3	0.1	0.0	0.1	0.4	0.7	1.2	0.0	0.0	0.7	0.5	0.2	0.1
	PrecStd	7.1	1.1	2.0	2.0	2.4	2.2	2.6	1.9	1.6	2.4	2.7	2.1	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	80.8	85.3	89.8	94.0	99.0	103.6	103.9	97.4	90.6	82.3	75.2
		MCWB	56.3	60.2	62.3	65.2	71.7	70.9	70.8	70.3	69.3	69.6	63.8	60.9
	2%	DB	71.6	74.9	80.5	83.8	90.1	95.5	99.6	100.0	94.4	87.0	77.3	69.8
		MCWB	54.1	58.9	62.2	65.4	70.7	71.4	70.9	71.1	71.1	68.5	61.6	57.0
	5%	DB	66.2	70.4	75.5	80.8	86.7	93.3	97.1	98.1	91.3	83.2	73.1	65.5
		MCWB	54.2	57.6	60.2	64.7	69.3	72.1	71.8	71.8	70.7	67.0	61.0	55.2
	10%	DB	62.7	66.1	72.3	77.3	84.0	90.8	94.7	95.4	88.9	80.4	70.0	61.3
		MCWB	52.3	56.3	59.1	63.4	69.0	71.6	72.2	72.3	70.0	64.7	59.4	52.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.2	66.1	67.8	71.2	75.8	78.0	77.8	77.3	76.2	73.6	69.5	65.0
		MCDB	66.8	73.4	77.9	79.6	87.2	90.4	91.0	91.2	87.1	82.2	76.8	69.6
	2%	WB	60.1	63.7	65.8	69.3	73.5	76.1	76.3	76.0	74.9	72.2	66.0	60.5
		MCDB	65.2	69.6	74.5	78.4	83.9	87.8	89.4	89.6	85.1	80.9	71.1	65.3
	5%	WB	56.6	61.0	64.0	67.8	72.2	74.8	75.3	75.1	73.7	70.8	63.9	57.8
		MCDB	64.2	67.1	70.9	76.2	82.0	86.1	88.3	88.4	83.6	79.1	69.6	62.4
	10%	WB	52.0	57.8	62.4	66.3	70.9	73.6	74.3	74.2	72.8	68.9	61.5	53.7
		MCDB	62.0	64.5	69.3	74.1	80.1	85.0	87.0	87.3	82.7	76.8	68.4	60.4
Mean Daily Temperature Range	5% DB	MDBR	19.3	19.4	20.0	20.0	18.4	18.6	19.2	19.4	18.9	19.8	19.3	18.2
		MCDBR	24.5	24.5	23.7	23.1	21.3	20.8	21.2	21.3	21.1	21.3	22.0	23.3
		MCWBR	14.3	13.7	11.3	11.3	8.1	5.8	5.4	5.3	6.2	9.2	11.6	13.8
	5% WB	MCDBR	21.8	22.2	18.8	19.2	18.2	18.8	18.9	18.8	17.5	18.7	18.5	19.6
		MCWBR	15.9	15.6	10.7	11.5	8.2	6.9	5.9	5.5	6.1	9.4	13.6	15.5
Clear-Sky Solar Irradiance	taub	0.303	0.314	0.346	0.378	0.414	0.427	0.429	0.425	0.389	0.350	0.324	0.308	
	taud	2.508	2.486	2.416	2.350	2.313	2.311	2.345	2.337	2.384	2.480	2.491	2.493	
	Ebn at Noon	291	298	295	289	278	272	272	271	277	282	280	282	
	Edh at Noon	27	30	35	39	41	41	40	39	36	30	27	26	
All-Sky Solar Radiation	RadAvg	926	1104	1445	1753	1888	2119	2148	1969	1642	1283	983	808	
	RadStd	95	172	104	145	194	202	146	118	148	191	128	99	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page